Biodiversity Loss and UK Beef Imports:

Is biodiversity loss in beef-exporting countries associated with the volume of beef exports to the United Kingdom?

A Cross-National Analysis

Raw data files are in data/raw/ (bio-raw.csv, FAOSTAT_data_en_3-31-2026.csv).
The complete analytical pipeline is documented in data/code/exploratory_report_bii_uk_beef_exports_sk7.ipynb.
Intermediate and final datasets are saved in data/clean/.

1. Introduction

Global agricultural expansion is among the leading drivers of biodiversity loss, particularly through land-use change associated with livestock production. Agriculture occupies approximately 43% of the Earth's ice-free and non-desert land surface and is responsible for around 31% of global greenhouse gas emissions, making it the dominant driver of terrestrial environmental change (Poore and Nemecek 2018). Within this system, livestock production—and beef production in particular—exerts a disproportionate environmental footprint: despite providing only 17% of global calories and 38% of protein, animal agriculture uses 83% of global farmland, primarily due to the large land requirements for grazing and feed production (Ritchie and Roser 2024).

Beef production is especially land-intensive. Cattle require significantly more area than other food sources, both directly for grazing and indirectly for feed crops. Expanding livestock systems have been linked to reductions in species abundance and the disruption of ecosystems, particularly in biodiverse regions such as the Brazilian Amazon (Garcia et al. 2017). At the same time, increasing globalisation has produced complex international supply chains in which environmental impacts are geographically separated from consumption—a phenomenon described as the externalisation of environmental costs (Wiedmann and Lenzen 2018). Recent work on deforestation embodied in international trade has demonstrated that consumer countries can drive substantial habitat conversion in producer nations without bearing the ecological consequences domestically (Pendrill et al. 2019).

The United Kingdom is a significant importer of agricultural products, including beef (GOV.UK 2025). This raises the question of whether patterns of UK consumption are associated with biodiversity outcomes in exporting nations. The present study investigates the research question: *Is biodiversity loss in beef-exporting countries associated with the volume of beef exports to the United Kingdom?* To examine this, country-level biodiversity data, measured using the Biodiversity Intactness Index (BII[1]), are combined with bilateral trade data on beef exports to the UK. By constructing both panel datasets and cross-sectional averages, the study evaluates whether higher export volumes are systematically associated with greater biodiversity loss.

2. Data and Methodology

2.1 Data Sources

The analysis integrates two primary datasets. The first is the BIO dataset (`bio-raw.csv` in `data/raw/`), which provides the Biodiversity Intactness Index (BII) sourced from the PREDICTS database (Newbold et al. 2015). The BII measures the average proportion of original species abundance remaining in a given area, with values ranging from 0 (completely degraded) to 1 (fully intact). The analysis utilises historical BII data from 2001 to 2014 and projected data from 2015 to 2020 under the SSP2 scenario[2] (Riahi et al. 2017).

[1]The BII measures the average abundance of native species in a given area compared to their abundances in an intact, undisturbed ecosystem. Values range from 0 (completely degraded) to 1 (fully intact).

[2]SSP2 (Shared Socioeconomic Pathway 2) represents a "middle of the road" scenario in which social, economic, and technological trends do not shift markedly from historical patterns (Riahi et al. 2017).

The second dataset comprises bilateral trade statistics from FAOSTAT (`FAOSTAT_data_en_3-31-2026.csv` in `data/raw/`), specifically annual quantities of "Beef and Veal" exports, measured in tonnes, from global reporting countries to the United Kingdom. The trade data cover 2001 to 2024. All cleaning and merging steps are documented in `data/code/exploratory_report_bii_uk_beef_exports_sk7.ipynb`, Sections 1–4. Intermediate outputs are saved to `data/clean/`.

2.2 Data Preparation

Country harmonisation. Country names differ between the BIO and FAOSTAT databases (for instance, FAOSTAT records "China, mainland" and "Netherlands (Kingdom of the)"). A custom mapping dictionary was developed to convert all reporting entities to ISO 3166-1 alpha-3 codes[3]. After mapping, 100% of FAOSTAT rows corresponding to UK imports were successfully matched (see notebook Section 2c). The overlap between the two datasets yielded a working sample of countries present in both sources.

Variable construction. A biodiversity loss variable (`bii_loss`) was constructed as 1 – BII, so that higher values indicate greater ecological degradation. This transformation ensures that a positive regression coefficient intuitively represents more exports being associated with more environmental damage. Trade volumes were transformed using ln(1 + x)—the natural logarithm, not log base 10—to reduce the influence of extreme exporters and approximate a more symmetric distribution suitable for linear modelling (raw skewness was substantially reduced after transformation; see notebook Section 3b).

Sample filtering. Countries with total exports below 10 tonnes across all years were dropped to focus on economically meaningful trade relationships, leaving a primary sample of 32 exporting nations (see notebook Section 3c for the full list of excluded countries). The primary panel covers 2001 to 2014, using exclusively historical BII data (`panel_2001_2014.csv` in `data/clean/`). An extended panel incorporating SSP2 projections (2015–2020) is used for supplementary analysis (`panel_extended_2010_2020.csv`).

2.3 Variance Decomposition

Before selecting model specifications, the total variance of each key variable was decomposed into between-country and within-country components (notebook Section 6). This step is critical for model selection because Fixed Effects (FE) estimators rely exclusively on within-country variation to identify effects. If a variable changes almost entirely across countries rather than within them over time, FE will have minimal identifying variation.

Table 1. *Variance Decomposition of Key Variables (Primary Panel, 2001–2014)*

Variable	Overall Var.	Between (%)	Within (%)
bii_loss	0.02	100%	0.4%
log(UK import qty)	9.74	88%	9%

Exact variance figures are available in the notebook's Section 6 output. Percentages reported here are rounded from the computed decomposition.

[3]ISO 3166-1 alpha-3 codes are standardised three-letter country codes (e.g. BRA for Brazil, ARG for Argentina) used to harmonise country identifiers across international databases.

The decomposition reveals that more than 99.5% of BII variation occurs between countries, with virtually no change within countries over the 14-year panel. This has two implications. First, cross-sectional OLS (which exploits between-country differences) captures the dominant signal and directly addresses the stated research question. Second, panel FE—which strips away all between-country variation via entity dummies—has almost no remaining variation in `bii_loss` to identify an effect. The FE estimator therefore tests a different, much narrower question: do countries that experience tiny year-to-year BII changes also see correlated export changes? This motivates presenting cross-sectional OLS as the primary specification and FE models as robustness checks.

2.4 Model Specifications

Five model specifications are estimated to examine the relationship from multiple perspectives. The primary specification is a cross-sectional OLS model on country-level means (2001–2014), estimated with HC3 robust standard errors[4]:

$$ln(1 + Export\ Volume) = \alpha + \beta \times bii_loss + \varepsilon$$

where the log-transformed mean annual beef export volume to the UK is the dependent variable and mean `bii_loss` is the independent variable. The use of country-level averages smooths volatile annual fluctuations in trade to capture the structural, longer-term relationship between a nation's typical export profile and ecological state.

Additionally, four panel models are estimated on the full 2001–2014 panel with entity-clustered standard errors: Pooled OLS, entity Fixed Effects, two-way Fixed Effects (entity and time), and Random Effects. A Mundlak (Correlated Random Effects) specification[5] decomposes the effect into within-country and between-country components, providing a formal test of whether these two dimensions of association differ in sign or magnitude. Model selection is guided by three diagnostic tests: the poolability F-test (H_0: all entity effects = 0), the Hausman test[6] (FE vs. RE), and the variance decomposition presented above.

3. Results

3.1 Descriptive Statistics

The primary panel comprises 32 countries observed over 14 years (2001–2014). Initial exploration reveals substantial variation in both trade volumes and ecological conditions. Trade data exhibits pronounced right-skewness: a small number of dominant exporters (notably Ireland, the Netherlands, and Brazil) account for the vast majority of UK beef imports, while most countries export relatively small quantities. The BII also varies considerably across exporters, with `bii_loss` ranging from approximately 0.05 (relatively intact ecosystems) to above 0.60 (substantial degradation). Summary statistics, top/bottom exporters, and distributional plots are reported in the notebook's Sections 5a–5d. Of particular

[4]HC3 robust standard errors (MacKinnon and White 1985) provide consistent inference in the presence of heteroscedasticity, where the variance of residuals may differ across observations.

[5]The Mundlak (1978) approach adds the group mean of the independent variable as an additional regressor in a Random Effects model, decomposing the total effect into within-group and between-group components.

[6]The Hausman test evaluates whether the more efficient Random Effects estimator is consistent. Rejection ($p < 0.05$) indicates that unobserved country-level characteristics are correlated with the independent variable, favouring Fixed Effects.

note is the heatmap of BII loss by country and year (notebook Section 5d), which visually confirms that almost all colour variation runs between rows (countries) rather than across columns (years)—a pattern consistent with the variance decomposition in Table 1.

3.2 Cross-Sectional OLS (Primary Result)

The cross-sectional OLS model, estimated on country-level means with HC3 robust standard errors, yields the following result:

Table 2. *Cross-Sectional OLS Results (HC3 Robust Standard Errors)*

Variable	β	SE	p-value	R^2	N
bii_loss	7.19	3.32	0.030	0.12	36

*** denotes significance at the 5% level. Full regression output including standard errors and confidence intervals is available in the notebook's Section 7b.*

The estimated coefficient of 7.19 indicates that a one-unit increase in biodiversity loss is associated with a 7.19-unit increase in the natural log of UK beef import quantity. Since the dependent variable is log-transformed, this represents a proportional increase in export volumes: countries with higher biodiversity loss tend to export substantially larger volumes of beef to the UK. The relationship is statistically significant at the 5% level ($p = 0.030$). The R^2 value of 0.12 indicates that biodiversity loss explains approximately 12% of the cross-country variation in export volumes—a modest but meaningful share given the single-predictor specification.

The scatter plot with ISO3 labels and fitted line (Figure 1 in notebook Section 7c; see `data/code/exploratory_report_bii_uk_beef_exports_sk7.ipynb`, cell 27) visualises this relationship. The upward slope is broadly distributed across the sample rather than driven by a single cluster of countries.

3.3 Panel Model Comparison

To examine whether the cross-sectional finding is robust to alternative specifications, four panel models were estimated on the full 2001–2014 dataset. The results, along with the cross-sectional OLS for reference, are presented in Table 3.

Table 3. *Model Comparison: Coefficient on bii_loss Across Specifications*

Model	β	SE	p-value
Cross-sectional OLS (HC3)	7.19	3.32	0.030
Pooled OLS	7.85	3.39	0.021
FE (entity)	−22.1	9.46	0.020
FE (entity + time)	-21.3	11.3	0.061
Random Effects (clustered)	-0.64	3.84	0.866

*** $p < 0.05$. n.s. = not significant. Full standard errors, confidence intervals, and observation counts are available in the notebook's Section 8e*

The most notable finding is the sign reversal between cross-sectional and FE specifications. While cross-sectional OLS and Pooled OLS both produce positive, significant coefficients (suggesting that countries with greater biodiversity loss export more beef to the UK), entity

Fixed Effects produces a negative and significant coefficient ($\beta = -22.15$, $p = 0.020$). This reversal is explained by the variance decomposition in Table 1: since over 99.5% of BII variation is between countries, FE—which identifies only off within-country variation—is estimating from near-negligible movements that are likely confounded by demand shocks, trade policy shifts, or measurement noise.

The diagnostic tests confirm this interpretation. The poolability F-test ($F = 76.59$, $p \approx 0$) strongly rejects Pooled OLS, indicating that country-specific effects are significant. The Hausman test[7] ($\chi^2(1) = 6.19$, $p = 0.013$) rejects Random Effects in favour of Fixed Effects. However, the Hausman result must be read alongside the variance decomposition: FE is preferred on consistency grounds, but it absorbs the vast majority of the BII signal via country dummies, leaving it with almost no power to detect the between-country pattern that is the subject of the research question. Cross-sectional OLS therefore remains the specification most directly relevant to the hypothesis.

3.4 Robustness

Outlier analysis. Cook's distance[8] was computed for each country in the cross-sectional sample. Three observations exceeded the 4/N threshold: Ireland, Brazil, and Sweden. Excluding these countries *strengthens* the result rather than weakening it ($\beta = 7.61$, $p = 0.005$), confirming that the finding is not driven by a small number of influential exporters (notebook Section 7d–7e).

Mundlak decomposition. The Mundlak (Correlated Random Effects) model[9] formally separates the within-country and between-country effects of `bii_loss`. Both components are statistically significant but opposite in sign: the within-country effect is negative ($p = 0.020$), while the between-country effect is large and positive ($\beta = 29.37$, $p = 0.003$). This confirms that the positive cross-sectional association reflects a genuine structural pattern across countries, while the negative FE result reflects a distinct (and much noisier) within-country dynamic.

Lagged specifications. Fixed Effects models with one-year and two-year lagged `bii_loss` were estimated to test whether prior biodiversity conditions predict current export volumes. Results are reported in the notebook's Section 9a.

Trade threshold sensitivity. The minimum trade threshold was varied from 10 to 50 to 100 tonnes. The FE coefficient on `bii_loss` remains stable across all thresholds (notebook Section 9c), indicating that the results are not sensitive to the exclusion of minor exporters.

Serial correlation. The Wooldridge test[10] for first-order serial correlation in FE residuals finds no evidence of autocorrelation ($p = 0.494$), validating the use of standard clustered standard errors (notebook Section 9d).

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[8]Cook's distance measures the influence of each observation on the overall regression. The standard threshold of 4/N identifies observations whose removal would substantially alter the fitted model.

[10]The Wooldridge (2002) test for serial correlation in panel data regresses first-differenced residuals on their lagged values and tests whether the coefficient equals −0.5, which would indicate no autocorrelation.

Table 4. *Combined Robustness Summary*

Specification	β	p-value	N
CS OLS (HC3) — primary	7.19	0.030	36
Pooled OLS (clustered)	7.85	0.021	339
FE (entity)	−22.15	0.020	339
Mundlak (within)	-22.15	0.020	339
Mundlak (between)	29.37	0.003	339
Wooldridge serial corr.	0.50	0.494	—

Exact coefficients and standard errors for all specifications are in the notebook's Section 9e.

4. Discussion

4.1 Interpreting the Cross-Country Pattern

The central finding is a positive, statistically significant association between biodiversity loss and UK beef import volumes at the cross-country level. Countries with greater ecological degradation tend to export larger volumes of beef to the United Kingdom. This pattern is confirmed by the cross-sectional OLS ($\beta = 7.19$, $p = 0.030$), the Pooled OLS ($\beta = 7.85$, $p = 0.021$), and the Mundlak between-country component ($\beta = 29.37$, $p = 0.003$), and it strengthens when the three most influential observations are removed ($\beta = 7.61$, $p = 0.005$).

The entity Fixed Effects model tells a different story, producing a negative coefficient ($\beta = -22.15$, $p = 0.020$). This apparent contradiction is resolved by the variance decomposition: less than 0.5% of BII variation occurs within countries over the 14-year panel. FE therefore identifies off movements too small to credibly isolate a meaningful effect. The negative within-country estimate likely reflects temporal noise, demand shocks, or minor measurement fluctuations rather than a true reversal of the cross-national pattern. The Mundlak model nests both signals and shows they are each statistically significant but operate at fundamentally different scales.

4.2 Model Fit and the Role of Omitted Variables

The R^2 of 0.12 indicates that biodiversity loss explains a modest share of export volume variation. This is expected rather than problematic: export volumes are shaped by many factors not captured in a single-predictor model, including comparative advantage, historical trade relationships, geographic proximity to the UK market, trade agreements, and domestic consumption patterns. The `bii_loss` variable is itself derived from national-level data that aggregates ecological conditions across entire countries, rather than isolating the specific regions where beef production occurs. This spatial mismatch (national BII versus production-region ecology) further limits the proportion of variance the model can explain.

The notebook's Section 10 outlines a framework for addressing omitted variable bias by incorporating controls such as agricultural land area, GDP per capita, and cattle herd size. This is flagged as future work pending data acquisition. Such controls would test whether the observed association persists after accounting for structural economic differences between countries.

4.3 Limitations

First, the analysis does not establish causality. The positive coefficient on `bii_loss` indicates association but does not demonstrate that UK demand directly causes biodiversity loss in exporting countries. Both variables may be jointly influenced by confounding factors such as agricultural expansion, land availability, or global commodity price dynamics. Causal identification would require instrumental variables or sub-national data that are not available in the current study.

Second, the `bii_loss` variable is a national aggregate that cannot capture sub-national land-use dynamics. Beef production in countries like Brazil is concentrated in specific regions (notably the Amazon frontier), while the BII is computed across the entire national territory. This spatial aggregation reduces the precision with which biodiversity loss can be linked to export activity.

Third, the extended panel (2015–2020) relies on SSP2 projections for BII, introducing model uncertainty. While this approach expands the dataset beyond the limited historical overlap period, the projected values depend on assumptions about future socioeconomic conditions that may not fully reflect actual developments.

Fourth, no control variables are included. GDP, land-use intensity, agricultural policy, herd size, and trade agreement status are all likely to influence both biodiversity loss and export capacity. Their omission creates the risk of omitted variable bias, and the coefficient should be interpreted accordingly.

Fifth, the 10-tonne minimum trade threshold removes approximately 22 countries from the sample, skewing the remaining 36 nations toward Europe and Latin America. This sample selection may limit the generalisability of the findings.

5. Conclusion

This study set out to test whether biodiversity loss in beef-exporting countries is associated with the volume of beef exports to the United Kingdom. Using a cross-sectional regression model based on country-level means over 2001–2014, the analysis finds a positive and statistically significant relationship ($\beta = 7.19$, $p = 0.030$) that is robust to outlier exclusion ($\beta = 7.61$, $p = 0.005$), trade threshold variation, and alternative model specifications. The Mundlak between-country component confirms the pattern at high significance ($\beta = 29.37$, $p = 0.003$).

The relationship is a between-country phenomenon, not a within-country dynamic. Entity Fixed Effects produce a negative coefficient, but the variance decomposition demonstrates that this reflects the near-total absence of within-country BII variation rather than a genuine reversal. The finding is therefore best understood as a structural cross-national pattern: countries that have experienced greater cumulative biodiversity loss tend to be larger beef exporters to the UK, consistent with the land-intensive nature of beef production.

The analysis does not establish causation, and the R^2 of 0.12 underscores that biodiversity loss is one factor among many shaping trade volumes. Future work should incorporate structural controls (agricultural land area, GDP, herd size), use sub-national BII and production data where available, and consider instrumental variable strategies to move toward causal identification.

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Appendix A: Project Structure and Reproducibility

All data and code are organised within the following directory structure. To reproduce the analysis, navigate to `data/code/` and open `exploratory_report_bii_uk_beef_exports_sk7.ipynb`. Execute all cells sequentially. The notebook reads from `data/raw/` and writes intermediate outputs to `data/clean/`.

```
project/
├── data/
│   ├── clean/                          ← Processed datasets
│   │   ├── bio_clean.csv
│   │   ├── bio_historical.csv
│   │   ├── crosssection_avg.csv            ← Country means for OLS
│   │   ├── fao_clean.csv
│   │   ├── merged_ols_ready.csv
│   │   ├── panel_2001_2014.csv             ← Primary panel
│   │   ├── panel_2010_2014.csv
│   │   ├── panel_extended_2010_2020.csv     ← Extended (with SSP2)
│   │   └── panel_regression_results.csv
│   ├── code/                           ← Analytical notebooks
│   │   ├── bio-sort.ipynb
│   │   ├── data_cleaning_and_preparation_sk7.ipynb
│   │   └── exploratory_report_bii_uk_beef_exports_sk7.ipynb  ← KEY
│   └── raw/                            ← Untouched source data
│       ├── bio-raw.csv
│       └── FAOSTAT_data_en_3-31-2026.csv
├── PCIRR-study-design-table-07.pdf
└── bio_diversity_report_sk7.pdf
```

Appendix B: Data Dictionary

Variable	Source	Description
bii	BIO	Biodiversity Intactness Index (0–1, where 1 = fully intact)
bii_loss	Derived	1 – BII; higher values indicate greater biodiversity loss

uk_import_qty_t	FAOSTAT	Annual beef export quantity to UK (tonnes)
log_uk_import_qty_t	Derived	ln(1 + uk_import_qty_t); natural log transformation
iso3	Derived	ISO 3166-1 alpha-3 country code for merging datasets
bio_source	Derived	Flag: "historical" (2001–2014) or "ssp2_projected" (2015+)

Appendix C: Notebook Section Reference

The exploratory notebook (`exploratory_report_bii_uk_beef_exports_sk7.ipynb`) is divided into the following sections, each of which corresponds to specific claims made in this report:

Section	Topic	Report Reference
1	Raw data inspection	Section 2.1
2	Country harmonisation	Section 2.2
3	Variable construction	Section 2.2
4	Merged panel construction	Section 2.2
5	Descriptive statistics	Section 3.1
6	Variance decomposition	Section 2.3, Table 1
7	Cross-sectional OLS	Section 3.2, Table 2
8	Panel models	Section 3.3, Table 3
9	Robustness checks	Section 3.4, Table 4
10	Control variable framework	Section 4.2
11	Visualisation summary panel	Figures referenced throughout